

Phase 2: Requirement Analysis

Streamlining IT Procurement: Automating Standard Laptop Orders with Flow Designer

Team ID	SWTID-2026-6800	Team Size	5
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1. Purpose

This phase lists what the automated laptop-order solution must do, the conditions it must work under, and the tools needed to build it.

2. Functional Requirements

ID	Requirement	Description	Priority
FR-1	Flow creation	Create a flow named "Standard laptop task" in Flow Designer, application Global, run as System User	High
FR-2	Service Catalog trigger	The flow starts from the Service Catalog trigger when a catalog item is ordered	High
FR-3	Create Catalog Task	Use the Create Catalog Task action, mapping the Requested Item record from the trigger to the Request Item field (table Catalog Task [sc_task])	High
FR-4	Task content	Set short description "Laptop needs to Configured" and the same value in the Description field	High
FR-5	Auto assignment	Set Assignment group to Hardware so the task lands in the hardware team queue	High
FR-6	Approval value	Set the Approval field on the catalog task to Approved	Medium
FR-7	Activation	Save and activate the flow so it runs when trigger conditions are met	High
FR-8	Catalog mapping	Attach the flow to the Standard Laptop catalog item (Maintain Items > Process Engine > Flow)	High
FR-9	Ordering and approval	Users can order Standard Laptop from Service Catalog > Hardware, and an approver can approve the request	High
FR-10	Traceability	REQ, RITM and SCTASK records must be linked and visible to the user and IT staff	Medium

3. Non-Functional Requirements

Category	Requirement
Usability	Ordering must follow the normal catalog experience; no extra steps for the requester
Reliability	Every approved standard laptop request must produce exactly one catalog task

Category	Requirement
Performance	The catalog task should be created within moments of the trigger, with no manual wait
Maintainability	Flow must be built visually in Flow Designer so any admin can read and change it
Security	Flow runs as System User; only authorised roles can edit flows and catalog items
Scalability	The same pattern can be reused for other standard hardware items
Auditability	Flow executions can be reviewed in Flow Designer's Executions tab

4. Data Flow Overview

Data moves through the solution as follows:

- 1 The requester opens *Service Catalog > Hardware > Standard Laptop* and clicks Order Now.
- 2 ServiceNow creates a Request (REQ) and a Requested Item (RITM) and raises an approval for the approver.
- 3 The Service Catalog trigger starts the flow "Standard laptop task" with the Requested Item record as data.
- 4 The Create Catalog Task action writes a new record into the Catalog Task [sc_task] table with the short description, description, Hardware assignment group and approval value.
- 5 The task appears under the Catalog Tasks tab of the RITM and in the Hardware group's queue.

5. Technology Stack

Component	Details
Platform	ServiceNow (Personal Developer Instance)
Automation tool	Flow Designer (Process Automation)
Trigger / Action	Service Catalog trigger; Create Catalog Task action (ServiceNow Core spoke)
Tables	sc_request, sc_req_item, sc_task, sys_user_group
Catalog item	Standard Laptop (Lenovo Carbon x1: Intel Core i5, 512GB SSD, backlit keyboard)
Group	Hardware assignment group
Access	Browser; admin role for build, requester and approver roles for testing

6. Assumptions and Constraints

- A ServiceNow instance with Flow Designer and the Service Catalog is available.
- The Standard Laptop catalog item and the Hardware group already exist.
- The flow is built in the Global application scope, as required by the project guide.
- The solution covers standard laptops only; other items and custom specifications are out of scope.
- Physical procurement, shipping and imaging are outside ServiceNow and are not automated here.

7. Outcome of the Phase

A complete and prioritised requirement list was prepared and will drive the design, development and test cases.